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A Qualitative Case Study: Exploration of Supply Chain Resilience in a Manufacturing SME

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Abstract

During recent years, disruptions in the supply chain, such as the COVID-19 pandemic, geopolitical uncertainties and increasing global complexity, have made organisations increasingly exposed to disruptions and uncertainty. For small and medium-sized enterprises (SMEs), this involves particular challenges due to less formalised processes and resource constraints. However, SMEs may benefit from informal routines such as flexibility, quick decision-making and close collaboration. Against this background, the study examines how a manufacturing SME develops and strengthens supply chain resilience, focusing particularly on organisational capabilities.

The study was conducted as a qualitative case study of the Swedish manufacturing company Tovenco, with empirical data collected through semi-structured interviews with employees from different parts of the organisation and analysed using thematic analysis.

The findings indicate that the case company attempts to create resilience through experience, flexibility and collaboration rather than through advanced strategies or technological systems. At the same time, internal organisational factors such as insufficient communication, unclear responsibilities, dependency on individual knowledge and limited use of digital systems create organisational vulnerabilities. The findings further show that the informal practices enabling flexibility and quick decision-making may simultaneously reinforce these vulnerabilities when responsibility and critical knowledge are concentrated among a few individuals. Therefore, the results highlight the importance of clearer structures for information sharing, knowledge distribution and planning.

The study contributes with empirical insights into how supply chain resilience is developed in a manufacturing SME, highlighting how characteristics that may strengthen resilience simultaneously create vulnerabilities.

Keywords: Supply Chain Risk Management, Supply Chain Resilience, Small and Medium sized enterprises, SMEs, Organisational Capabilities

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1. Introduction & Background

Over the past decades, global economic development has been increasingly transformed by liberalization in trade, technological innovation and improvements in transportation and communication (Slack et al., 2022). These developments have enabled companies to operate across borders and organize production on a global scale. As a result, supply chains have become increasingly complex and interdependent, increasing the need for organisations to maintain visibility, coordination and control across supply chain activities (Christopher & Peck, 2004; Shojaei et al., 2025).

In recent years, advances in digital technologies have continued to transform industrial operations. Technological developments have created new opportunities for firms to improve efficiency, optimize processes and use resources more effectively (Slack et al., 2022). While these changes have supported trade, innovation and economic growth, they have also increased interdependency within global supply chain networks. As a result, disruptions occurring in one part of the network can more quickly spread and affect other parts of the supply chain, thereby increasing vulnerability to unexpected disruptions (Shojaei et al., 2025).

The frequency of disruptions has also increased, with events such as natural disasters, pandemics and geopolitical tensions occurring more often. This has created additional challenges for companies, which highlights the importance of managing risks (Bak et al., 2020). Within the field of Operations and Supply Chain Management (OSCM) this has led to a growing focus on the concept of *Supply Chain Resilience*. Supply chain resilience can be defined as a supply chain's ability to resist disruptions, absorb shocks and recover from them. This by either returning to its previous level of performance, adapting operations to maintain performance or improving performance following a disruption (Christopher & Peck, 2004). However, despite existing guidelines, many firms struggle to implement resilience practices, particularly Small and Medium-sized Enterprises (SMEs) (Bak et al., 2020; Safari et al., 2024).

SMEs play a central role in both national and global economies and are deeply integrated into supply chains. Within the European Union, SMEs account for

approximately 99% of all businesses, highlighting their economic importance (European Commission, 2023). At the same time, SMEs often operate under more constrained conditions than larger organisations. Previous research therefore emphasises the need for SME-specific perspectives, as factors such as limited financial and technological resources, operational constraints, lower investment capacity and less formalised organisational structures shape how SMEs develop resilience and respond to disruptions (Munongo & Pooe, 2024; Bak et al., 2020).

However, SMEs also possess characteristics that can support resilience. Their smaller size can enable faster decision-making, better flexibility and closer relationships with partners, which can facilitate quicker responses to disruptions (Koporcic et al., 2025). Similarly, previous research highlights that proactive and innovative organisational cultures are positively associated with supply chain resilience in SMEs (Munongo & Pooe, 2024). This highlights the importance of examining how internal organisational capabilities influence the development of resilience in SMEs that are operating under resource-constrained conditions.

1.1 Case company

This study focuses on Tovenco, a Swedish family-owned SME that manufactures commercial kitchen ventilation systems and kitchen hoods. Tovenco fits the definition of a SME with approximately 20–30 employees, including both production workers and office staff (European Commission, n.d; Jimmy Åström, personal communication, 2026). The firm is divided into two segments, the private household segment and the commercial kitchen segment and serves a broad market consisting of construction firms, schools, restaurants, laboratories, large infrastructure contractors and private households. The company operates in a customised production environment that depends on the availability of metal components, motors, filters and other specialised parts (Jimmy Åström, personal communication, 2026). This reliance on externally sourced components increases the firm's vulnerability to supply chain disruptions, as even minor delays can interrupt production processes, delay deliveries and installations and create operational challenges across the organisation (Nakandala et al., 2025).

1.2 Relevance and contribution

Supply chain resilience is increasingly viewed as a strategic necessity for SMEs, since disruptions have become more frequent and severe (Sun et al., 2023). Tovenco's situation reflects several challenges commonly associated with SMEs, including dependence on a small number of suppliers, limited financial and organisational resources and a reliance on informal communication rather than formalised processes (Shojaei et al., 2025). These characteristics make Tovenco a relevant case for examining how supply chain resilience is developed and applied in an SME context.

Scientifically, this study contributes to discussions in supply chain management and resilience theory by examining how resilience is formed under resource-constrained conditions. Existing resilience literature often emphasises approaches such as structural redundancy, advanced technologies and formal risk management as important elements (Christopher & Peck, 2004). However, previous research states that many of these approaches may be more difficult for SMEs to implement due to limited financial, technological and organisational resources (Bak et al., 2020; Safari et al., 2024). This highlights a mismatch between dominant resilience frameworks and SME realities, which further emphasize the importance of examining how resilience is practiced within an SME context (Bak et al., 2020). This study contributes to research in SME resilience literature by providing an empirical example of how characteristics commonly described as resilience-building within SMEs, at the same time may contribute to vulnerabilities in practice.

1.3 Research gap

Although supply chain resilience in SMEs has received increasing academic attention, existing research remains largely focused on conceptual frameworks and quantitative relationships, comparatively fewer studies examine how resilience is achieved in everyday practice (Bak et al., 2020; Lopez et al., 2024). In addition, limited attention has been given to whether the capabilities commonly identified as resilience-building in previous literature actually function effectively in practice within SMEs.

Several studies identify internal capabilities such as flexibility, collaboration, digitalisation and proactive decision-making, as important factors for building resilience in SMEs (Bak et al., 2020; Koporcic et al., 2025; Munongo & Pooe, 2024). However, much of the existing literature focuses on what SMEs should do to become resilient rather than how these capabilities are implemented, adapted, and sustained in practice (Bak et al., 2020; Lopez et al., 2024). Consequently, researchers increasingly call for qualitative and empirically grounded studies that examine resilience processes and day-to-day resilience practices within SMEs (Nakandala et al., 2025; Koporcic et al., 2025; Bak et al., 2020; Lopez et al., 2024).

1.4 Purpose & Research Question

The purpose of this study is to investigate how a small manufacturing SME manages and approaches supply chain resilience under resource constrained conditions, with a broader focus on the role of internal capabilities. As there is still limited empirical understanding of how resilience functions in everyday SME practices, the following research questions were developed. The first research question constitutes the main focus of the study:

- *How do SMEs develop and strengthen supply chain resilience in practice with the resources they have?*

To answer this question, the study first examines the internal organisational capabilities that influence how disruptions are managed in practice. It then uses insights from previous research together with the empirical findings to reflect on potential areas where the case company could further strengthen its supply chain resilience. Together, the sub-questions provide a deeper understanding of how resilience is developed and how it may be improved within SMEs.

    - *What role do internal organisational capabilities play in shaping supply chain resilience?*
    - *How can insights from existing organisational practices be used to reflect on potential areas for strengthening supply chain resilience in SMEs?*

1.5 Scope and delimitation

By conducting a qualitative case study of a single SME, this study is able to examine how supply chain resilience is developed in practice and explore resilience processes in greater depth. A qualitative approach makes it possible to highlight organisational dynamics that may be difficult to capture through quantitative approaches (Bak et al., 2020). Rather than statistical generalisation, the findings are intended to provide analytical insights for SMEs operating in similar settings and to give context-specific insights into resilience in a manufacturing SME.

The study is based on a case study approach, where the empirical data consists of qualitative methods, primarily semi-structured interviews with employees from different parts of the case company's organisation. This approach allowed the study to capture multiple perspectives on the organisation's day-to-day practices.

The study concentrates on internal organisational practices, specifically those related to procurement, production planning, communication and decision-making. Although external factors such as supplier relationships, geopolitical events and broader supply chain disruptions are recognised as important, they are primarily examined based on their internal effects on the case company. The study therefore does not include perspectives from suppliers, customers or other external actors.

2. Theoretical Background

In order to build a conceptual background for the study, this chapter introduces key concepts and previous research related to supply chain resilience in SMEs. Based on the research topic and the variation in perspectives, relevant peer-reviewed articles were selected to offer different perspectives and a focused understanding of the topic. Together, the literature provides a nuanced understanding of how SMEs are affected by disruptions.

2.1 Supply Chain Resilience

Resilience can be understood as the ability to act or react to unpredictable events to minimise negative effects on organisational performance and profitability. This includes the ability to maintain an existing stable state or adapt to a new one, highlighting how organisations cope with change, absorb shocks and recover from disruptions (Bak et al., 2020).

Some studies interpret resilience primarily as an outcome, while others describe it as a range of internal and external capabilities. This variation demonstrates that resilience is a multifaceted concept in the SME context. Bak et al. (2020) suggest that SME resilience is shaped by the interaction between several dimensions, including collaboration, internal capabilities, information systems and financial conditions, rather than by one single dominant factor.

Therefore, resilience includes both reactive actions taken during disruptions and proactive efforts focusing on preventing and minimising risks (Koporcic et al., 2025). Key components within resilience often include flexibility and collaboration throughout the entire supply chain, as well as the ability to reallocate resources and adapt processes when needed (Christopher & Peck, 2004). These capabilities help organisations manage uncertainties and minimise the negative effects of disruptions. In addition, SMEs' adaptability and agile decision-making support both innovation and recovery in uncertain contexts. Innovation is therefore highlighted as an important mechanism through which SMEs strengthen their resilience by enabling adaptation of the business model, improvement of services and response to changes in the environment (Lopez et al., 2024).

However, resilience needs to be understood in the specific context of SMEs. Even if it is important for all organisations to achieve resilience, the way in which it is developed and maintained may differ significantly between larger organisations and SMEs. Compared to larger organisations, SMEs operate under more restricted conditions, including limited financial and technological resources, less formalised structures and broader environmental conditions. As a result, resilience in SMEs is context-dependent and differs significantly from the resilience of larger organisations (Bak et al., 2020).

2.2 Supply Chain Disruptions in SMEs

To better understand the challenges SMEs face in their work to achieve supply chain resilience, it is important to identify the types of disruptions that may occur. Disruptions can arise from different sources and affect different parts of the supply chain. This section categorises disruptions into internal and external disruptions and thereby creates a foundation for analysing how SMEs are affected.

2.2.1 Internal disruptions

Internal disruptions refer to issues that arise within the organisation and affect internal operations. These include production halts, machine failures, planning errors as well as organisational and management-related flaws. Such disruptions often arise from technical faults, inadequate processes or inefficient internal management. This can affect the firm's ability to maintain stable operations (Safari et al., 2024).

Internal disruptions may also arise as a result of implementing and utilising new technology. These challenges are often not only connected to the technology itself, but also to how the implementation process is managed within the organisation. In some cases, the implementation of a system is carried out without sufficient planning, clear guidelines or adequate training, making it difficult for employees to understand how the system should be used in practice. This can lead to resistance or uncertainty among users. Particularly as new technology generally demands changes in established ways of working, roles and responsibilities (Davenport, 1998). As a result, employees may use informal ways of working and work in parallel with other systems, which may reduce effectiveness and create inefficiencies within the organisation (Seo, 2013).

Furthermore, Safari et al. (2024) emphasise that internal capabilities such as knowledge management, information sharing and organisational structure are central to maintaining reliable and efficient supply chains. Deficiencies in these areas increase the likelihood of internal disruptions that affect operational performance. Additionally, Bak et al. (2020) highlight that the resource constraints commonly faced by SMEs, such as financial limitations, lack of time and limited personnel resources, can hinder the implementation of new and more efficient processes, thereby increasing vulnerability to internal disruptions.

2.2.2 External disruptions

External disruptions refer to situations that originate outside the organisation's control and affect both the business and the supply chain. These may include geopolitical events, supplier-related disruptions, pandemics such as COVID-19, inflation and other external economic uncertainties. The effects of disruptions are often significant for SMEs due to limited resources, dependency on external actors and low bargaining power within supply chain networks (Bak et al., 2020).

SMEs are particularly vulnerable to supplier disruptions, such as delivery delays and material shortages. Sun et al. (2023) highlight how shocks and disruptions can lead to shortages of critical resources, while Nakandala et al. (2025) explain how disruptive events may interrupt the flow of goods and services throughout the supply chain.

This vulnerability is further reinforced by the structural conditions commonly associated with SMEs. Limited resources and dependence on a small number of suppliers often reduce their ability to identify alternative solutions when interruptions occur. Furthermore, external disruptions are frequently systemic in nature, meaning that events occurring in one part of the world can generate global consequences throughout supply chains. For SMEs, this dependence on suppliers may lead to increased prices, limited resource availability and heightened uncertainty within the supply chain (Nakandala et al., 2025; Shojaei et al., 2025).

2.3 Building Supply Chain Resilience in SMEs

Building supply chain resilience is a complex process in SMEs and often differs from how resilience is developed in larger firms. Larger firms often have greater opportunities to implement approaches such as formal risk management systems, supplier diversification and structural redundancy to strengthen supply chain resilience. Previous research describes these approaches as important elements of supply chain resilience (Christopher & Peck, 2004). However, SMEs often develop resilience through a combination of internal capabilities and external relationships. Previous research therefore highlights different dimensions that affect how resilience is developed and maintained within SMEs. (Bak et al., 2020).

To provide a structured understanding of how resilience is developed within SMEs, this section presents a few of the central dimensions identified in resilience research. A preliminary literature analysis identified four recurring dimensions often discussed in previous research on SME resilience. The first one focuses on organisational capabilities and examines internal factors such as coordination, flexibility and decision-making. The second dimension addresses collaboration with suppliers and emphasises the importance of good relationships with external partners when managing disruptions. The third dimension relates to resource constraints and SME-specific conditions and emphasises how limited resources and structural characteristics affect resilience. The fourth dimension focuses on technology as a support for resilience and examines how digital tools make it possible for organisations to better manage and respond to disruptions. Together, these dimensions illustrate how SMEs develop resilience in practice.

2.3.1 Organisational Capabilities

Resilience in SMEs is shaped by internal organisational capabilities, information systems and financial conditions, all of which may limit SMEs' ability to prepare for and manage disruptions (Bak et al., 2020). At the individual level, resilience is often supported by strong leadership, fast decision-making and managerial control. This appears particularly common in family-owned SMEs, where decision-making is often concentrated to a limited number of individuals (Koporcic et al., 2025). At an organisational level, capabilities such as adaptability, organisational culture, effectiveness and strategic responsiveness are important

to enable organisations to manage crises and changing conditions (Koporcic et al., 2025). These capabilities make it possible for SMEs to identify, utilise and restructure existing resources when disruptions occur. Previous research further shows that resilience in SMEs is often embedded in everyday organisational practices and operational experience rather than in highly formalised systems and structures (Koporcic et al., 2025).

In this context, proactive resilience can further strengthen organisational capabilities by enabling SMEs to balance efficient use of existing resources and routines with the ability to develop new solutions and respond to changing conditions (Pertheban et al., 2023). However, several factors may simultaneously limit SMEs' ability to effectively prepare for and manage disruptions. Information- and communication-related challenges, including weak coordination, limited information sharing and lack of trust and commitment appear to be particularly important (Mahmud et al., 2021).

2.3.2 Supplier relationships and collaboration

Supply chain collaboration is often highlighted as an important driver of performance and a central part of resilience in SMEs. Collaboration is also described as an important way to overcome structural and relational constraints, rather than solely relying on internal capabilities (Mahmud et al., 2021). Organisational resilience is therefore influenced by external connections and interactions between different organisations and other actors. SMEs can increase resilience by utilising collaboration with other SMEs in a complementary way and by aligning resources within the network (Koporcic et al., 2025).

Similarly, access to support and resources from partners can contribute to strengthening the organisation's ability to improve resilience and manage disruptions more effectively (Nakandala et al., 2025). Close relationships with suppliers and customers can help compensate for limited internal resources, while collaboration enables knowledge sharing and more effective responses. At the same time, SMEs have limited influence over partners in the supply chain, making collaboration a factor both enabling resilience and a potential source of vulnerability (Bak et al., 2020).

This makes supplier relationships and proximity to suppliers a key factor for resilience in supply chains, especially within SME contexts. This highlights the importance of close relationships, even if the effectiveness may vary depending on the organisation and the level of disruptions they face (Nakandala et al., 2025). Altogether, SMEs are highly dependent on effective coordination, information sharing and close supplier collaboration in order to manage disruptions and maintain supply chain resilience.

2.3.3 Resource constraints and SME-specific conditions

Large enterprises often have greater opportunities to manage supply chain risk better than SMEs in many ways, this by relying on capabilities such as structured routines, planning and information programs. SMEs, however, often need to focus on a more agile approach with greater sensitivity to the business environment and may not be able to invest in such programs, even though they may be crucial for their resilience (Safari et al., 2024).

In addition, SMEs have a tendency towards short-term planning and reacting to circumstances as they arise. Decisions in SMEs are often managed by the owner or top management, which can create centralised control within the organisation. Although this can enable fast decision-making and quick responses to disruptions, it is not always sufficient for creating long-term resilience. Even though such strategies can be difficult to implement in SMEs due to less formalised structures, previous research suggests that such ways of working are often important for strengthening resilience over time (Halkos et al., 2018).

Furthermore, SMEs often face underlying structural challenges, such as limited ability to prepare for disruptions, which affect their ability to recover and restore profitability within the organisation (Bak et al., 2020). Therefore, internal challenges such as resource constraints and management perspectives are highlighted as important factors influencing resilience (Halkos et al., 2018). In addition, these limitations can also make continuous learning, innovation and improvement work more difficult, despite these being central factors for strengthening resilience over time (Bak et al., 2020).

2.3.4 Technology as a Support for Resilience

Digital technologies can strengthen supply chain resilience through tools that improve visibility, transparency and real-time information sharing across the supply chain. By monitoring and managing supply chains through real-time digital technologies, SMEs may also respond more quickly to unexpected disruptions and changes in the business environment (Shojaei et al., 2025).

A commonly used technological tool in manufacturing businesses is an Enterprise Resource Planning (ERP) system. This system can support coordination and information sharing within the organisation by integrating different functions, such as purchasing, production and inventory management into one system. ERP systems may support faster and more effective decision-making during disruptions by centralising information, standardising processes and improving visibility. This ability can enable organisations to identify problems earlier, coordinate responses and maintain operational continuity (Shojaei et al., 2025; Mathew & Johansson, 2023). In this regard, ERP systems can contribute to supply chain resilience by supporting organisational coordination, reducing information gaps and improving visibility when implemented and used effectively.

ERP systems often include planning functions such as Material Requirements Planning (MRP), predefined lead times and structured Bills of Materials (BOM). All of which aim to create stability and control in production and delivery processes (Monk & Wagner, 2008). Furthermore, ERP systems are used to develop and execute production plans and schedules in manufacturing environments (Mathew & Johansson, 2023).

However, despite the potential benefits of ERP systems, research shows that several challenges limit the success of their implementation. Coşkun et al. (2022) identify a large number of so-called Critical Failure Factors (CFF), factors that are recurring causes of failure in ERP implementations. These factors are not only technical, they also include organisational aspects, such as weak governance, communication problems, insufficient training and poor data quality. Therefore, ERP implementations are often complex and difficult to manage and failures are rarely explained by a single cause. Problems tend to arise from a combination of

factors that affect the organisation's ability to use the system efficiently (Coşkun et al., 2022). Adopting technology can therefore be challenging due to limited access to skilled employees, incompatible technologies and weak supply chain networks (Mahmud et al., 2021).

3. Methodology

The development of supply chain resilience in SMEs is influenced by many connected factors and is built through everyday work, interactions and informal practices. Since these processes are typically context-dependent and influenced by employees' experiences and organisational interactions, they can be difficult to fully understand using only surveys or experiments (Bak et al., 2020). Given the exploratory nature of the research question, which seeks to understand how supply chain resilience is developed and enacted in practice, a qualitative case study approach was seen as appropriate. This approach is particularly suitable for examining processes, experiences and meanings within real-world contexts (Creswell & Creswell, 2023; Eisenhardt, 2021). The choice of a case study is further supported by Yin (2014), who argues that case studies are well suited for examining complex situations within real organisational contexts. In addition, existing research on SME supply chain resilience has largely relied on surveys, modelling and conceptual frameworks, which limits insight into everyday practices (Nakandala et al., 2025; Munongo & Pooe, 2024). By examining a single SME in depth, the study enabled a detailed understanding of how internal capabilities and supplier relationships interact under resource constraints.

3.1 Data collection

To collect data for this study, both a literature study and empirical data collection were conducted. The literature study, presented in the theoretical framework, served as a foundation for the study by providing an overview of existing research on supply chain resilience in SMEs. The aim was to understand how current research conceptualises and defines SME supply chain resilience, what types of strategies or solutions have been proposed and which research gaps have been identified in the literature. This also supported the development of the interview guide and helped ensure that relevant information was covered in the empirical data collection.

The primary data consisted of semi-structured interviews based on open-ended questions, which allowed the respondents to explain their experiences in their own words, while also giving the researcher flexibility to explore emerging themes and probe deeper into relevant issues (Creswell & Creswell, 2023).

3.1.1 Literature study

The literature used to support this study was collected from several sources, including academic journal articles and books. The literature study was conducted using a structured process that started with identifying relevant keywords, for example: “supply chain resilience”, “SMEs”, “external disruptions”, “internal capabilities”, “supplier collaboration”, “ERP systems” and “risk management”. Thereafter, systematic searches were conducted in academic databases such as Web of Science, Google Scholar and Scopus. The initial search resulted in 26 relevant articles. After screening the articles based on relevance to the study’s aim, research questions and SME context, 16 articles were selected for the final literature review.

Articles were excluded if they focused primarily on large organisations, industries outside the scope of the study, or topics not directly related to supply chain resilience and internal organisational capabilities within SMEs. However, where relevant, earlier foundational studies were included to provide theoretical grounding.

The literature was then organised into key thematic areas, meaning that recurring patterns and central concepts across the studies were identified and grouped. This approach enabled the identification and summarisation of the most relevant contributions within the field (Creswell & Creswell, 2023; Braun & Clarke, 2006).

3.1.1.1 Analytical Framework

The analytical framework of this study is inspired by previous research on SME supply chain resilience, particularly Bak et al. (2020), which highlights resilience as shaped through the interaction between organisational capabilities, collaboration, information systems and resource conditions. Through the literature review, four analytical dimensions were identified and used throughout the study; organisational capabilities, supplier relationships and collaboration, resource constraints and technology as a support for resilience. The identified dimensions align with those identified by Bak et al. (2020) to a large extent, which further supports the relevance of the analytical framework in relation to existing SME resilience research.

3.1.2 Interviews

The selection of respondents for the interviews was made with purpose and aimed to capture a broad range of perspectives within the organisation. A total of seven employees were interviewed, which represents approximately one-third of the workforce at Tovenco. To ensure that the collected data was relevant to the research questions, employees with direct involvement in supply chain-related activities, were prioritised. The respondents represented different organisational functions, including production, procurement, planning, administration and management, with varying levels of experience, ranging from approximately 2 to 40 years. This enabled the study to capture both strategic and operational perspectives on how supply chain disruptions are managed in practice, while also giving insights into the organisation's different operational challenges, disruptions and changes over time.

The interview guide consisted of open-ended questions in different areas: disruptions, everyday practices, internal organisational capabilities, resource constraints, supplier relationships, proactive strategies and overall reflections. Emphasis was placed on capturing detailed and context-specific insights and each main question was complemented with probing questions (See Appendix 1). This allowed answers to move beyond general descriptions and instead capture real situations. In addition, the interviews supported an understanding of internal organisational capabilities, as well as how external relationships and resource constraints can influence the firm’s ability to respond to disruptions. The semi-structured format provided flexibility for the interviewers to adapt probing questions depending on the respondents answers, while ensuring consistency across interviews.

To ensure equal time for each participant, a predefined timetable was used to schedule all interviews. Each respondent was allocated approximately 30 minutes to ensure sufficient depth in the collected material. Six interviews were carried out on-site at the case company, while one interview was conducted remotely. The interview process involved two researchers. During the interviews the roles were clearly defined, with one conducting the interview, while the other was responsible for managing the audio recording and taking supplementary notes.

All interviews were fully transcribed for the purpose of making the analytical material available for in-depth analysis.

3.2 Data analysis

The interview data was analysed using qualitative coding and thematic analysis. The transcribed interviews were reviewed several times to gain an overall understanding of the material. Relevant sections related to supply chain disruptions and resilience practices were identified and initially coded. These codes were then organised and developed into the four overarching dimensions identified from literature. Within each dimension, sub-codes were created to capture more specific aspects of the data and to group related experiences and statements from the respondents (See Appendix 2). In addition, to identify similarities and differences between the respondents' answers, selected quotes and examples were used to illustrate patterns in the data (Braun & Clarke, 2006).

This approach provided a structured analysis while capturing the organisational context in which resilience practices took place. The analysis was guided by existing theoretical concepts, such as internal capabilities, relational resilience and resource constraints, while also allowing identification of new insights that emerged from the interviews.

3.3 Research quality and ethics

Several measures were taken throughout the study to ensure research quality and ethical integrity. To avoid selective interpretation of the empirical material, efforts were made to maintain objectivity during both data collection and analysis by collecting data as accurately and comprehensively as possible (Saunders, 2019).

Given the qualitative case study design and focus on a single organisation, the findings are not intended to be statistically generalisable. Instead, the study aims to provide analytical insights into how supply chain resilience is developed and managed within a manufacturing SME operating under resource-constrained conditions. The findings may therefore be relevant for SMEs facing similar challenges, such as limited resources, dependence on key individuals, close supplier relationships and relatively informal organisational structures. However, transferability should be considered with caution, as factors such as industry

characteristics, organisational size, ownership structures and supply chain complexity may influence how resilience is developed and managed in other contexts. The findings should therefore be understood as context-specific insights rather than conclusions applicable to all SMEs.

An important approach that was used throughout this qualitative study was triangulation, which was applied to ensure credibility and to reduce bias (Donkoh & Mensah, 2023; Noble & Heale, 2019). Investigator triangulation was applied through the involvement of two researchers during both the interviews and the analysis. This contributed to reducing interpretation bias and enabled a more nuanced understanding of the empirical material (Noble & Heale, 2019). In addition, empirical findings were continuously compared with previous literature and theoretical concepts, which further supported the triangulation process (Donkoh & Mensah, 2023).

A potential source of bias identified in the study is that respondents in managerial positions may have had an interest in presenting the organisation in a more positive manner. To reduce this risk, interviews were conducted with employees from multiple functions and in different organisational levels. To ensure a nuanced understanding of the organisation, the study included respondents with differing experiences, perspectives and roles. This reduced the risk of relying on a single viewpoint (Noble & Heale, 2019).

Additionally, credibility was strengthened through the use of semi-structured interviews, probing questions and transparent documentation of the analytical process. All participation in the study was voluntary and all participants were clearly informed about the purpose of the study and their role within it. Transparency was supported by describing to respondents how data would be collected and interpreted. By removing identifying information from the empirical interview data, the study ensured anonymity and confidentiality. All data were stored securely and used only for academic purposes. These practices align with established research ethics principles including: informed consent, participant's right to withdraw from the study and the protection of privacy and confidentiality (Saunders, 2019).

3.4 Use of AI

In the writing of this thesis, ChatGPT has been used as a tool for improving grammar, spelling and language clarity. It was also used for suggesting alternative wording, translating words and phrases to ensure appropriate academic language as well as assisting with the overall structure of the thesis. All prompts were formulated to ensure that no new information was introduced. All suggestions were critically reviewed and revised by the authors.

4. Empirical Findings

This section presents the empirical findings from the conducted interviews. The findings are structured according to the key themes identified in the empirical data. To ensure anonymity, the interview participants are referred to as Respondent 1 to Respondent 7 throughout the findings section.

4.1 Interview Findings

The findings from the interview transcripts were structured into four key themes that derived from the thematic coding (see Appendix 2), which also align with the key dimensions identified in the theoretical framework. The aim of the interviews was to get an insight into how supply chain resilience is enacted in everyday practices in the case company based on the experience of the respondents.

4.1.1 Organisational capabilities

Several internal organisational capabilities emerged as important in the organisation's handling of supply chain disruptions. A recurring theme throughout the interviews was the organisation's flexibility in responding to disruptions, where problems are often solved in a situation-adapted manner. Although knowledge and responsibilities appear highly dependent on specific individuals, respondents describe how the employees collaborate to manage disruptions as they occur.

Respondent 5 described how disruptions overall are *"handled case by case"*, while Respondent 4 explained that *"we just keep going and deal with things as they come"*. At the same time, respondents describe a strong willingness to support each other across roles and adapt their work to quickly resolve disruptions as they arise. This is illustrated by Respondent 2, who highlighted how employees often help each other across functions when problems occur:

> *"I can go from the office to the production and ask them to explain things to me, so that I can then explain it to the customer. [...] In some way, that is quite unique, that we are able to solve many things like that. Customers receive a lot of support here, and I believe that is very important as well."*

Most respondents described how disruptions are handled based on previous knowledge and individual competence, where decision-making largely relies on

experience and intuition. Despite this, the interview findings reveal differing perceptions regarding how structured the organisation is and whether clear processes are actually in place. Respondent 6 described the structure in the organisation as:

> *"We have processes that are used, although they are not as developed and formalised as in larger companies, but we do have processes."*

Several other respondents expressed a more critical view of the organisational structure. Respondent 3 described the organisation as:

> *"No structure, it is the spinning wheel that controls things there [...] we are basically shooting from the hip."*

Respondent 4 also highlighted a lack of structure in purchasing routines:

> *"No structure. It is more like, 'oh, it has been a long time since we ordered filters, let's order some."*

Respondent 1 explained that the organisation often ends up "doing a lot of firefighting." Further describing that the current way of working enables fast decision-making, but also results in a low degree of standardisation.

In addition, respondents describe a general willingness to collaborate internally, while also expressing a need for more structured forms of communication. For example, Respondent 5 describes a limited use of planning routines and coordination mechanisms:

> *"There are no meetings or gatherings where we discuss things, solve problems or make plans. Communication is the biggest issue, because everyone is very skilled at doing their own tasks, but there are many communication failures."*

This is further illustrated by Respondent 4, who explained that:

> *"We don't really have any structured meetings, we just talk when needed."*

Although respondents describe how disruptions are often managed collaboratively through informal ways of working, the interviews also reveal several

organisational challenges related to maintaining an overview during disruptions, as well as a lack of shared goals due to limited coordination and communication.

Respondents also describe how responsibilities and competence are distributed across individuals within the organisation. Absence, such as illness, affects the organisation's operations, as certain tasks and knowledge are linked to specific individuals in the production. Respondent 5 explained that:

> *"If someone on the shop floor is absent due to illness, the entire chain can come to a halt."*

Similarly, Respondent 2 noted that:

> *"There is only one person responsible for each position in the production [...] It is very important that everyone is here."*

Respondents linked this to limitations in time and resources, which makes it difficult to train employees to perform additional tasks or to hire additional staff.

Leadership and coordination represent another factor. Several respondents described how decision-making is concentrated among top management and that responsibility is rarely delegated. Respondent 1 explained that "one person should not be involved in everything, because then nothing gets done," further arguing that responsibility should be delegated to a greater extent throughout the organisation. Respondent 5 further emphasised this:

> *"I think there is also a tendency for management to want control over many parts of the operations, making it difficult to fully delegate responsibilities. A large amount of time is spent on operational tasks in production, while more administrative and strategic activities sometimes receive less attention [...] a clearer distribution of responsibilities and giving production staff more responsibility could possibly free up more time for needed work instead."*

This statement is further reinforced by Respondent 1:

> *"There is a feeling that they need to have control over everything, instead of giving someone responsibility for the entire workshop and then letting*

that person manage it. It might work for a short while, but then someone quickly steps in and interferes again."

4.1.2 Supplier relationships and collaboration

This section refers to how respondents perceive the organisation's ability to maintain strong relationships with their suppliers and how these relationships influence operational performance. All of the respondents described how relationships with suppliers play a central role in how disruptions in the supply chain are handled in practice. Several of them emphasised how close relationships with suppliers support the firm, through faster feedback, greater flexibility and a stronger willingness from suppliers to prioritise the firm's orders. Respondent 7 described "*it's important to have a good relationship with critical suppliers, the ones who really matter,"* and noted that problems are often more easily solved with these suppliers.

Respondent 1 described how suppliers sometimes can provide early warning signals, for example by informing the firm about longer delivery times or potential future risks. This gives the company the ability to adapt their planning to place orders earlier. For example, Respondent 1 described how a supplier provided early indications of an upcoming shortage during COVID-19:

> *"It was the absolute biggest disruption. Luckily, we received indications about it very early [...] they told us that we should probably start thinking about how to solve the situation and they also said that there was really only one way to become independent from the problem: to build up a huge inventory."*

The respondents mentioned that Tovenco discussed this indication internally and made the decision to purchase a huge inventory. According to the same respondent, this enabled them to build up inventory for 1.5 years ahead and continue delivering as usual and thereby avoid major disruptions.

The importance of receiving early indications from external actors was further supported by Respondent 7, who stated:

> *"We also receive some information from our suppliers, the bank and other contacts, which can sometimes provide early indications that something is about to happen. If you maintain regular contact with suppliers, they are also quite good at informing you in advance."*

At the same time, the respondents explained that dependence on specific suppliers or components creates vulnerability. In one case, Respondent 6 specifically described how a supplier's bankruptcy resulted in Tovenco needing to find alternative suppliers. As a result, they now always have backup suppliers for critical components. This was supported by several respondents, who stated that the company actively works on identifying alternative suppliers for critical items.

Respondent 1 stated that they rarely experience problems with suppliers and that issues can almost always be resolved. They also noted that when problems did occur, they were often due to internal factors at Tovenco, such as placing orders too late. Respondent 5 provided an example involving an order for filters for the commercial kitchen hoods. The respondent explained:

> *"I was very stressed and was about to send an order for filters [...] then I realised that I had ordered the wrong filters. A huge quantity and also the wrong amount. By then, they had already started producing the filters, but they stopped the production and solved it. It resulted in a few extra days of delivery time, but they solved it. So that supplier is really good."*

The respondents also highlighted that communication is an important factor during disruptions. They further indicate that close communication is important to obtain information about delivery status, discuss solutions and sometimes to coordinate actions. However, the respondents also noted that communication sometimes varies, since some suppliers can be difficult to reach and that can make it more challenging to manage disruptions. Respondents 1 and 4 described shifting from suppliers in distant countries to suppliers closer to Sweden to reduce risks:

Respondent 1:
"A major reason why it works so well is that we have moved almost everything back home. When I say 'home', I mean that we now purchase the materials in Sweden. Previously, our sourcing was spread across several European countries and China, but now we have moved almost everything back home."

Respondent 4:
"We have moved many suppliers closer, from Poland and the Czech Republic and places like that, I think."

This in order to avoid disruptions related to events such as war or pandemics.

Coordination with suppliers at Tovenco, is often carried out with the aim of minimising risks. This, through dialogue about forecasts, planning and inventory management. The level of structure in this work varies, but several respondents stated that it to a large extent is based on ongoing contact and established relationships rather than formalised processes.

4.1.3 Resource constraints and SME-specific conditions

The respondents describe how resource constraints influence both the organisation's ability to manage disruptions and its opportunities to work more proactively with supply chain resilience. Limited resources appear to affect several parts of the organisation, including inventory decisions, staffing, planning and long-term development initiatives. A recurring aspect identified throughout the interviews concerns financial and capital-related constraints. For example, Respondent 6 states that *"one does not want to tie up capital,"* referring to how financial limitations affect inventory decisions.

This approach is reinforced by Respondent 3, who describes how a large Mall project resulted in significant financial losses due to errors in production. The Mall project required extensive rework, where large parts had to be replaced due to faults, leading to substantial costs tied up in inventory before payment was received. This severely strained the organisation's financial position and they were denied governmental financing, resulting in layoffs of several employees. This

experience created a setback, which continues to influence Tovenco's current decision-making and inventory. Additionally, respondent 7 described how external disruptions, such as the war in Ukraine and the blockage of the Suez Canal, led to component shortages and longer lead times.

Respondents also acknowledge a need for more structure, while describing time constraints as the recurring limitation. They state that efforts to improve processes are affected by limited time and personnel. As described by respondent 6:

> *"Right now, we are not fully able to keep up because we are growing and lack the resources that would be required. We could become much bigger with the snap of a finger and under the right conditions, but at the moment the development happens more gradually because we lack resources."*

Time constraints as a limitations is further described by respondent 7:

> *"You simply get far too little time for each individual detail. If you had more time, you could do a better job as well [...] documentation is also important, because if things are documented properly, it becomes much easier to go back and find information later."*

Respondent 5 explained that *"everything is managed on a day-to-day and hour-to-hour basis,"* illustrating how operational work often takes priority over long-term development and improvement efforts. Similarly, Respondent 2 described how *"the development takes a lot of time [...] that time is difficult to find here,"* despite recognising the need for more structure and organisational development.

Several respondents also described difficulties in completing development initiatives and formalising structures. Respondent 2 explained that

> *"they never really finish building things [...] the product comes out on the website before everything around it is completed."*

The respondent further described how *"there becomes a lot of 'made-up solutions' that are not really finished,"* reflecting how limited time and resources contribute to reactive and incomplete processes.

In addition, respondents describe challenges in allocating time for development activities and in covering for absences. Respondent 2 stated that *"it is very important that everyone is here,"* while Respondent 1 argued that *"we would probably need one person dedicated only to these questions"* regarding resilience work. They also describe how decisions related to expansion, inventory levels and resource use are managed within existing constraints.

4.1.4 Technology as a support for resilience

Digital systems and technology emerge quite prominently in the interviews. The company uses an ERP system called Monitor ERP. This system is described as a central tool for the company and it is used in their supply chain operations. Although the system is not currently utilised to its full potential, several respondents emphasised that they believe a more structured and integrated use of the ERP system could significantly improve the organisation's internal processes. In particular, respondents describe how the system could improve planning, inventory management, traceability and information flow. Thereby creating better conditions for coordination and proactive decision-making during disruptions. Respondent 7 explained that:

> *"We probably only use around 30 percent of the system. [...] It is a large and extensive system with many functions that could make things much easier, but we have not reached that point yet."*

Similarly, Respondent 1 described how the current lack of system adaptation contributes to manual routines and operational delays:

> *"I do not know Monitor particularly well, but it is a business system that could provide a lot of value. [...] Many of our purchasing routines become manual. That is our absolute Achilles heel. Sometimes we fall behind with purchasing and this leads to delays in deliveries."*

The respondents further connect the ERP system to the organisation's ability to maintain control over inventory and production flows. Respondent 3 explained that:

> *"Monitor is an enormous support tool if the inventory was managed correctly. [...] If we had continuous inventory management every week so that everything actually matched, we would also be able to see what will be needed. [...] But if inventory is not managed correctly [...] then we are essentially not using the function properly."*

Furthermore, Respondent 5 provided an example of how deficiencies in inventory management affect the functionality of the ERP system. The respondent explained that the system contains information about products and uses BOM structures to define components, but that some components do not have article numbers and are therefore not properly registered in inventory:

> *"Like this lamp for example. It was the first time I sold it, but it is included in another product. It had broken at a customer site, so we sent a new one. It exists within the product in the system, but not as a separate item. But then I also do not think that the lamp in the ERP-system is connected to the product in a way that deducts it each time the product is sold. The lamp itself is purchased separately, and then the product is assembled here. Some parts are missing.*
> *[...]*
> *There is a lot of that. Pricing is also a major issue here, since prices are not set for many products, meaning that we often have to calculate prices manually and ask around."*

According to the respondent, when a product is produced, the inventory balance is not updated accordingly. This leads employees to track inventory separately and that the ERP system does not reflect actual stock levels. Respondents further describe that not all items are correctly registered in the system and that inventory is not continuously updated. It also emerges that there are no clear routines for goods receiving and inventory control, which affects the overall

visibility of material flows. Similar challenges were also described regarding pricing processes, where employees often need to manually calculate prices and request information from others due to incomplete or missing data in the ERP system.

A lack of training in the system is also evident in the interviews, with several respondents stating that they have never received formal training and have instead been self-taught or learned from colleagues. It also emerges that not all employees use the system to the same extent, which means that information in some cases is handled and communicated manually between colleagues. For example, respondent 4 described that employees in production do not use the system and that it is mainly limited to certain office staff.

Furthermore, Respondent 7 stated that system competence has been lost over time, since much of the knowledge in the ERP-system was previously held by a former employee. This employee drove the implementation and understood how the system functioned and built the foundation for the system. When this person left the company, the effective use of the system declined, as no one fully knew how to manage it. Respondent 7:

> *"Monitor works well and everything is there, but we are not fully trained in how to use it. A previous employee was very skilled and when he left, we lost a lot of knowledge [...] Now, many things have become manual again, which is unfortunate since the system is advanced and expensive. The plan is to clean up and learn how to use it properly again."*

Several respondents describe how they instead rely on manual solutions as a complement to the system, such as Excel or other self-developed workarounds. Excel is used for calculations and pricing, particularly in cases where the ERP system does not contain updated or sufficient information. Respondent 6 further stated that there is a resistance to change within the organisation. Some employees who have worked at the company for many years are very reluctant to change. This reluctance is especially connected to increased workload or when broader skills are required. Respondent 4 added that:

“The problem is that many employees, at least within the commercial kitchen division, can barely log into a computer.”

4.2 Empirical Synthesis

The results from the interviews show that the disruptions faced by Tovenco in the supply chain are mostly connected to external factors, according to the respondents themselves. They mention events such as COVID-19, the blockage of the Suez Canal and Russia’s war in Ukraine as global disruptions in the transport flows, all of which have affected the company’s access to materials and lead times.

At the same time, internal factors are also highlighted. Production halts, such as machine stops, are not frequent and are not considered a problem by the respondents. Instead, internal challenges connected to planning, communication, leadership and the use of technological systems are highlighted. As well as resource constraints, such as limited time, employees and capital.

5. Discussion

The aim of this section is to analyse how supply chain resilience is developed in practice within an SME context, as well as to examine how the empirical findings align with or differ from the literature. The discussion is structured around key dimensions identified in both the literature and the empirical data and concludes with recommendations based on the authors' reflections and existing research in order to address the study's research questions.

This section also discusses how the studies included in the literature review relate to the resilience in SMEs. It is important to consider contextual differences that may influence the applicability of the literature. Since previous research is conducted across different regions, factors such as culture, institutional conditions, market structures and resource availability may influence how these challenges are described and affect SMEs in practice. However, the analysis of the case company shows several similarities to how SMEs and resilience practices are described in previous research. This suggests that the literature can still function as a relevant analytical reference frame for understanding how resilience is developed within SMEs.

5.1 Discussion of findings

The following sections discuss the findings in relation to the key dimensions, with particular focus on how these dimensions influence the development of supply chain resilience within the case company.

5.1.1 Organisational Capabilities

While disruptions are often initially understood as external events, this case study illustrates that resilience is not only determined by the disruptions themselves, but also by the organisation's ability to interpret, coordinate and respond to them. External events such as the COVID-19 pandemic, the war in Ukraine and the blockage of the Suez Canal all created huge pressure on the organisation. However, the findings indicate that the consequences of these disruptions were strongly influenced by the organisation's level of preparedness and internal capabilities.

During the COVID-19 pandemic, for example, Tovenco received early warnings from suppliers regarding potential material shortages, which enabled the organisation to secure inventory before the disruptions fully affected operations. According to the respondents, this helped reduce the overall impact of the pandemic on the organisation. In contrast, internal disruptions connected to, for example, the large Mall project eventually led to sudden layoffs, denied governmental funding and has continued to affect the organisation's current cautious mindset. Unlike the disruptions connected to external suppliers, this situation originated internally within the organisation.

The analysis therefore suggests that disruptions were managed more successfully when early information and external support enabled proactive actions for Tovenco. Whereas, internally generated disruptions exposed limitations related to a lack of structure, coordination and preparedness. Indicating that the organisation struggled to manage disruptions independently, particularly in situations where external support was not available.

This also appears connected to how resilience is developed within the organisation. While Tovenco's informal ways of working may support continuity during disruptions, they also seem to create uncertainty, operational stress and vulnerability (Bak et al., 2020; Safari et al., 2024). This further appears related to how organisational knowledge and responsibilities are currently distributed.

In the case company, this appears to be a significant challenge. Although the respondents described several valuable organisational capabilities, the analysis indicates that there is no overall structure connecting these capabilities in a systematic way. The organisation possesses a large amount of knowledge and competence. However, this knowledge is neither consistently shared or supported through broader responsibility distribution across the organisation. As a result, opportunities to strengthen long-term resilience become limited.

The organisation's development over time seems to have affected how resilience is achieved. Informal methods have gradually evolved alongside the company's growth rather than through strategic planning. The organisation's current ways of working may function while the company remains small. However, if the business

expands and operations become more complex, their informal approaches will become difficult to maintain. Expansion is something that the company has expressed a desire to pursue and the lack of clear structures may therefore make it difficult to support the growth they are aiming for. This shows a broad challenge often found within SMEs, where organisational growth often develops faster than internal coordination and communication structures (Bak et al., 2020; Safari et al., 2024).

Another challenge identified in the analysis is communication within the organisation. Much of the communication takes place informally through direct contact between employees, which can support flexibility in daily work. At the same time, important information is not always shared across the organisation. This creates difficulties in coordination, especially during disruptions. Since disruptions are handled differently depending on the individual, consistency in operations seems to vary and opportunities for shared learning across the organisation become more limited.

Furthermore, the analysis highlights how centralised decision-making both strengthens and limits resilience. In line with research on family-owned businesses (Brenes et al., 2011), decision-making and responsibility are concentrated among very few individuals in the case firm. Although this may sometimes create flexibility and faster processes, it instead appears to reduce delegation, slow down organisational development and limit the use of employees' knowledge and perspectives in the firm. Several of the organisation's ways of working appear to rely more on individual experience than on shared routines or documented knowledge. A significant risk to resilience is having too much knowledge dependency on certain individuals (Bak et al., 2020). When individuals are absent or leave the organisation, important knowledge and experience risk being lost. As a result, the organisation functions as long as the right individuals are in place, but this creates vulnerability when staff are sick or turnover occurs.

5.1.2 Supplier relationships and collaboration

Within the case company, close and long-term relationships with suppliers appear to be an important central factor for handling disruptions and

uncertainties in the supply chain. The organisation appears to create flexibility and adaptability even in situations where external conditions are rapidly changing, which aligns with previous research emphasising the importance of continuous dialogue, information sharing and collaboration for strengthening supply chain resilience in SMEs (Nakandala et al., 2025; Mahmud et al., 2021). Therefore, supplier relationships are not only important for daily operations, but also function as an important part of the organisation's overall resilience capability.

At the same time, the analysis indicates that these close relations also create new forms of vulnerability. When collaboration is concentrated to a limited number of suppliers, a dependency arises, making the organisation sensitive to disruptions among specific suppliers. This became particularly visible in situations where disruptions among individual suppliers directly affected operations, such as when a supplier bankruptcy forced the organisation to rapidly identify alternative sourcing solutions. This illustrates a central tension within SME-resilience, where close relations both strengthen and limit organisational flexibility. Although long-term collaborations increase trust, fast communication and shared problem solving, they may at the same time limit the organisation's flexibility if suppliers are lacking. This is especially clear in SMEs, where resources to diversify the supplier base often are limited (Bak et al., 2020).

In addition, the analysis highlights that the organisation seems to have a conscious and proactive approach to external risks after the recent years global disruptions. One practical way to reduce exposure to disruptions is to move suppliers geographically closer in order to create greater control over the supply chain, with shorter lead times and fewer transportation routes that may be affected by disruptions (Christopher & Peck, 2004). This is something Tovenco has done and previous experiences appear to have influenced this decision. Supplier resilience therefore seems to have developed through experience and adaptability rather than through a formalised strategy.

This further indicates that the analysis highlights that close relations and continuous communication with suppliers are an important tool for handling uncertainties. As illustrated by the COVID-19 example in the case company, early information about potential disruptions appears to create opportunities to act

proactively by adapting inventory management or planning before the disruption affects operations. This shows how information sharing acts as a form of organisational resilience in practice.

Overall, the analysis shows that the work with supplier relationships appears to be one of the areas where the organisation has come furthest with their resilience work. Their way of working with supplier relationships is characterised by continuous development and adaptability. This may indicate how smaller companies may have the opportunity to develop these close relationships, since suppliers may represent one of only a few external actors and therefore naturally become an important part of the organisation. However, although this way of working has functioned well so far, it may become increasingly important to develop more structured ways of working as the business grows.

5.1.3 Resource constraints and SME-specific conditions

The analysis further demonstrates how resource limitations shape the organisation's capabilities within the case company. At Tovenco, economic considerations appear to influence many organisational decisions, particularly regarding inventory management and long-term investments.

The findings suggest that the primary challenge is not necessarily a lack of capital itself, but rather uncertainty regarding where investments should be made. Although the firm describes its financial position as stable, the organisation still operates with limited financial resources, particularly in comparison to larger firms. As a result, there is a clear need to avoid tying up excessive amounts of capital, especially since the materials used within the business are relatively costly and can rapidly increase inventory value. This reflects a common challenge in SMEs, since they often lack the same opportunities as larger organisations, to build safety stock or invest in redundancy (Shojaei et al., 2025; Bak et al., 2020). Measures aimed at strengthening resilience also appear difficult to prioritise within the organisation. Consequently, investments that do not generate immediate operational or financial value seem to be deprioritised.

In this context, Tovenco's inventory management illustrates a tension between efficiency and resilience. Low inventory levels may increase vulnerability during

disruptions since there is no safety buffer, however, it also reduces tied-up capital and financial risks. Therefore, resilience is not only about eliminating risks, but rather about balancing different risks against each other and prioritising which are the most important to address. Tovenco may have to accept some operational vulnerabilities to maintain financial stability, which further strengthens the perception that resilience in SMEs differs from larger organisations in how they handle resilience.

Furthermore, the analysis indicates that lack of time may constitute an even greater constraint than lack of capital. Financial resources influence the organisation's ability to invest in long-term improvements. However, the main challenge appears to be to create space for development work within daily operations. Current operational tasks and problems are prioritised in the daily work and there is no time for developing strategic initiatives. This may contribute to reinforcing Tovenco's current reactive mode of working, since there simply is no time for proactive measures.

As previously mentioned, the concentration of responsibility among a few individuals appears to be an important reason for this, since tasks that could potentially be delegated instead remain with top management. If responsibilities had been distributed more broadly, there would likely have been more time available for improvement work, which does not necessarily have to rest only on top management. By involving employees from different functions Tovenco could also improve coordination, information sharing and broader organisational understanding, which previous research identifies as important aspects for resilience in SMEs (Bak et al., 2020; Mahmud et al., 2021).

5.1.4 Technology as a support for resilience

A recurring theme across all interviews is the respondents' dissatisfaction with the current business system, an ERP system that appears to complicate work processes rather than fulfil its intended purpose of creating visibility, coordination and structure. Although ERP systems may not typically be described as the main factor that creates resilience, the respondents strongly emphasise the system as an important way forward for the organisation. According to the respondents, a more structured use of the system could reduce manual work, improve inventory

accuracy and strengthen the organisation's ability to anticipate and manage disruptions more proactively.

Although technology itself does not automatically strengthen an organisation's resilience, its effectiveness largely depends on how well it is integrated into organisational routines and daily work. Tovenco's ERP system illustrates a clear difference between technology's potential value and its actual use in practice. Their system was implemented as a strategic investment to improve planning, coordination and information flows. However, the analysis shows the difficulties for the case company to fully integrate the system into daily work.

Coşkun et al. (2022) research on ERP implementation highlights how technological problems most often are closely connected to organisational and human factors. Poor data quality, unclear ways of working, limited user involvement and insufficient training together appear to have limited Tovenco's function and use of the system. Problems connected to data quality appear as particularly critical, since incorrect or incomplete information affects planning, inventory management and coordination between functions.

The analysis also suggests that the organisation's way of working is characterised by fragmented information flows and a lack of shared routines regarding system use. This indicates that the organisation has not yet established a shared digital way of working, where the ERP system acts as a central part of the operations. This emphasises the importance of coordination, information sharing and standardised processes for digital tools to be able to support organisational stability and resilience (Safari et al., 2024; Shojaei et al., 2025). In this case, however, the use of parallel solutions and manual ways of working appears to limit the system's potential to create visibility and coordination. Manual ways of working appear to persist because employees lack sufficient knowledge in how to operate the system correctly. This creates a vicious cycle where employees continue relying on manual solutions instead of fully integrating the ERP system.

The analysis further shows that resistance to change appears to affect the organisation's ability to use the system consistently. Parts of the organisation continue to work according to earlier routines, even when changes have been

proposed. This creates a situation where information becomes fragmented and where the full potential of the system to support decision-making and coordination is limited. In the analysis it became apparent that limited training and understanding of the system's function contribute to employees mainly using the parts of the system they feel comfortable with. What likely appears to be lacking in this case is insufficient governance and managerial support during the implementation. Successful ERP implementation requires commitment, clear coordination and consistent ways of working across the organisation to maintain continuity and reliability in the information shared within the system (Davenport, 1998; Coşkun et al., 2022).

In addition, system knowledge within the organisation also seems to be unevenly distributed. Knowledge about the system appears to a large extent to have been concentrated to one individual, creating both key competence being lost and vulnerability when this individual left the organisation. This indicates that technology may be difficult to sustain as a long-term organisational tool without continuous knowledge sharing, training and organisational engagement (Coşkun et al., 2022). The system's limited use is therefore not primarily a technical failure, rather a result of limited organisational anchoring and a lack of common ways of working with the system.

The analysis of Tovenco's use of ERP clarifies that resilience is not created through technological investments alone. The important part is the integration of technology into the organisation's structures, routines and competence throughout the organisation. Based on the shortcomings identified and how the employees expressed themselves regarding the ERP system, there appears to be significant potential for improvement, where not necessarily the ERP system itself, but rather the information and structure provided by the system when used correctly, could increase the case company's resilience long term.

The findings indicate that better use of the ERP-system could improve several operational areas that may indirectly strengthen resilience over time. Respondents repeatedly connect the system to improvements in inventory management, MRP, BOM structures and production planning on the shop floor. The system could also be used as a way to create clearer communication and

documentation, where information becomes more visible and less dependent on verbal communication between employees. Further strengthening the possibility for routines and ways of managing disruptions to be documented and used for future learning and organisational improvement.

Although the system appears to be a good fit for the organisation, there may potentially be other systems that are better suited in certain aspects. However, the current ERP system is highly versatile and Tovenco has already invested substantial time and financial resources into its implementation. The analysis further shows that investments in education, system integration and organisational development could make it possible for Tovenco to make better use of the system. These improvements would likely require additional investments and effort, but the long-term benefits would probably outweigh the costs that Tovenco currently seems hesitant to invest in.

5.2 Recommendations

Based on the analysis across the dimensions, there appears to be a clear need for more structured practices throughout the organisation. Although Tovenco demonstrates several characteristics commonly associated with resilience in SMEs. These capabilities do not appear to be the result of a consciously developed resilience strategy, but rather an outcome of daily problem-solving. The findings therefore also indicate several areas where the organisation could strengthen its resilience capabilities.

A central recommendation for the case firm across all dimensions would be to include employees to a greater extent in the organisation's operations and development work. Several of the challenges identified in the study seem connected to what the organisation describes as time constraints. There appears to be limited time available for long-term development initiatives and resilience work, since operational tasks and ongoing problem-solving take priority. To address this, some of the tasks that top management currently struggles to prioritise could instead be delegated to other employees. This could be done in order to improve the efficiency of development work and free up time for more strategic initiatives. A broader distribution of responsibilities and clearer role allocation could thereby create better opportunities for the organisation to

strengthen resilience over time. More employees could also be involved in tasks outside their primary responsibilities, thereby increasing shared organisational knowledge and reducing dependency.

The absence of strategies currently makes it difficult for employees to understand how their daily tasks contribute to the organisation's long term goals, which affects their sense of purpose. Therefore, if the organisation develops and communicates a clear strategy it's likely that employees could create a stronger sense of context and purpose. Such an approach could strengthen organisational alignment and improve the organisation's ability to manage disruptions. This may further improve coordination and organisational alignment during disruptions (Bak et al., 2020; Safari et al., 2024).

One way of working with this would be to introduce meetings, something the organisation lacks entirely. This could create a clear overview of tasks, priorities and current challenges, which in turn could improve coordination and reduce misunderstandings. These meetings could also function as two-way communication, where information would not only be shared from management to employees, but also from employees back to management. This could help ensure that employees' experiences, ideas and perspectives are also taken into consideration in the organisation's development work. Furthermore, this recommendation could also create better conditions for distributing responsibilities and tasks currently handled within top management. Previous research emphasises the importance of efficient information sharing and coordination for improving the organisation's ability to handle disruptions (Bak et al., 2020; Mahmud et al., 2021).

Furthermore, there appears to be a need to work more systematically with both the ERP system itself and the organisational routines connected to it. Conducting a more thorough review of the system together with the system supplier, improving data, inventory management, quality, ensuring complete BOM structures and clarifying how information should be registered could improve the system's functionality significantly. More structured training, clearer system guidelines and broader organisational competence regarding system use could further reduce dependency on manual work and parallel solutions.

The organisation would also benefit from distributing system competence more broadly throughout the organisation rather than concentrating knowledge to a few individuals. Stronger governance and clearer organisational expectations regarding how the system should be used may also reduce resistance to change and create better organisational integration of the system.

The analysis also highlights that information sharing with suppliers already functions as an important strength in the organisation's resilience work. However, the recommendation is to develop simpler models to evaluate supplier risks, for example based on criticality, geographical location, lead times and access to alternative suppliers. The organisation could also try to minimise its dependency on individual suppliers and work more with alternative suppliers or dual sourcing for critical components. This could both minimise the consequences of delivery problems and external disruptions, while also strengthening flexibility.

By working actively with these measures, the organisation would likely become more robust and resilience could become integrated in their daily operations. In addition, conditions for long-term development can be strengthened and continuity can be maintained despite employee absence.

5.3 Discussion Synthesis

The results of this study contribute to a deeper understanding of how supply chain resilience is developed in SMEs. . In addressing the main research question:

- *How do SMEs develop and strengthen supply chain resilience in practice with the resources they have?*

The findings show that the organisation's resilience is primarily developed through situation-adapted ways of working embedded in daily operations. At first glance, the company's practices appear to align relatively well with how resilience in SMEs is described in previous literature. Within the organisation, many of the characteristics commonly associated with resilience are clearly present.

At the same time, the findings raise questions regarding what resilience actually means in practice within SMEs. Although previous literature identifies several factors associated with resilience, the findings from this case study indicate that these do not always function as resilience-building in practice. Instead, several aspects described as resilience-enabling simultaneously appear to create organisational vulnerabilities within the case company. Rather than functioning as opposites, resilience and vulnerability appear closely interconnected within the organisation.

This also highlights how resilience within SMEs may differ from how resilience is often described in larger organisations, where resilience is more commonly associated with formal structures, strategic planning and systematic risk management. While Tovenco's current ways of working appear functional under existing organisational conditions, the organisation appears operationally adaptable rather than structurally resilient. Without more stable organisational structures and broader knowledge distribution, the organisation's current resilience capabilities may become increasingly difficult to maintain over time.

In relation to the first sub-research question:

- *What role do internal organisational capabilities play in shaping supply chain resilience?*

The findings show that internal organisational capabilities are central to how the organisation handles disruptions when they occur. Flexibility, experience and close collaboration enable the organisation to continue operating despite uncertainty and disruptions.

For example, flexibility and fast decision-making enable quick problem-solving and operational continuity, while simultaneously limiting the development of clear structures, shared routines and long-term coordination. Similarly, the organisation's dependence on experienced employees enables rapid responses during disruptions, while also creating vulnerability when knowledge and responsibilities are concentrated to a limited number of individuals. Direct and

informal communication further supports fast responses, but also increases the risk that information is not consistently shared across the organisation.

Much of the company's work appears to be largely based on "firefighting", where problems are solved as they arise rather than through proactive planning and clear structures. Fast decision-making and direct communication support short-term problem solving, but also contribute to inconsistent information sharing and informal working methods within the organisation.

The analysis also gives the impression that the organisation in practice is larger and more complex than its internal ways of working are adapted for. The company has high turnover, a fairly high level of production activity and appears to have the financial opportunities to expand and invest in improvements. At the same time, respondents repeatedly describe lack of time, structure and competence as central barriers to development. Production appears to be running constantly, while much of the organisation's work is characterised by continuous problem solving and operational pressure. The organisation therefore appears trapped in a reactive way of working, making it difficult to create space for long-term development initiatives.

Responsibility and decision-making are highly concentrated to a limited number of individuals, while there appears to be a reluctance to fully delegate responsibility to other employees. This may partly be understood in relation to previous research on family-owned firms, where centralised decision-making and strong managerial control are common characteristics (Brenes et al., 2011). In many ways, the organisation appears to be something that management holds very closely, where operations are largely driven through personal control and involvement rather than through shared structures.

The findings also show that technology and ERP systems play a more complex role than described in previous literature, at least within this specific SME. The analysis reveals a clear gap between the system's potential and its actual use within the organisation. This indicates that technological investments alone do not strengthen resilience, but rather that their effectiveness depends on

organisational structure, knowledge sharing and consistent system usage throughout the organisation.

Overall, the findings suggest that the organisation does not necessarily possess a particularly sustainable form of resilience. Instead, the organisation appears to have developed its own adaptive way of handling disruptions. In this way, the findings contribute to SME resilience theory by illustrating that practices commonly associated with resilience do not necessarily strengthen resilience in practice, but may simultaneously create organisational vulnerabilities within SMEs.

Finally, the findings, together with the recommendations presented throughout the discussion, contribute to answering the second sub-research question:

- *How can insights from existing organisational practices be used to reflect on potential areas for strengthening supply chain resilience in SMEs?*

The findings show that the organisation's current ways of working create both opportunities and limitations for further development. Several of the study's recommendations therefore focus on creating clearer organisational structures, broader responsibility distribution and improved knowledge sharing throughout the organisation. The organisation would likely benefit from creating more space for strategic improvement work and reducing its dependence on specific individuals. In the long term, there appears to be a need to develop the organisation's internal structures at the same pace as the organisation's size and complexity continue to grow.

6. Conclusion

This study has examined how supply chain resilience is managed within a manufacturing SME operating under resource-constrained conditions. The

findings show that the organisation primarily handles disruptions through flexible and experience-based ways of working embedded in daily operations, rather than through formalised structures or strategic resilience practices.

Although many of the characteristics observed within the organisation align with previous literature, the findings also reveal a more complex situation. The identified characteristics, such as flexibility, fast decision-making and informal coordination appear to support operational continuity during disruptions. However, the same characteristics also seem to create vulnerabilities, such as dependency on individuals, lack of structure and reactive ways of working. The conclusion of this study therefore suggests that resilience and vulnerability may exist simultaneously within SMEs rather than representing separate conditions.

The findings further highlight the importance of internal organisational conditions in shaping how disruptions are managed in practice. Factors such as communication, responsibility distribution, knowledge sharing and organisational structure appear highly influential in determining the organisation's ability to respond to uncertainty and operational challenges. The study also indicates that technological systems may support long-term resilience when combined with shared routines, organisational engagement and consistent system usage.

Overall, the study contributes to a more nuanced understanding of supply chain resilience in SMEs by illustrating how the characteristics often associated with resilience may simultaneously support short-term adaptability, while also contributing to long-term organisational vulnerabilities. The case company may appear resilient because it continuously manages to solve problems and maintain operations. The organisation's focus on short-term problem solving appears to limit opportunities to develop more stable structures and a stronger long-term organisational foundation.

6.1 Future Research

To better understand the structural conditions and their effect on resilience practices, future research could examine how resilience is developed in different SMEs and contexts. This could provide broader insight into whether the conclusions in this study are similar or differ across organisations, by conducting

comparative studies including multiple case companies. Such research could further examine the relationship between resilience and vulnerability in greater depth and how current informal practices may simultaneously create dependencies and operational challenges over time.

Additionally, future research could further contribute to the understanding of whether characteristics earlier identified as resilience-building actually function in practice for SMEs. More empirical research is needed to examine whether the resilience practices commonly described in previous literature actually function effectively in practice within SMEs, as well as to identify alternative approaches SMEs may use instead. Methods such as observations during a longer period could provide a deeper understanding of resilience in daily work. Since disruptions are often unpredictable and resilience is developed continuously, observations of daily work could contribute with concrete empirical examples.

Finally, there is a need for further research on how SMEs can integrate technology and digital systems into organisations in a way that supports both flexibility and long-term resilience.

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8. Appendices

Appendix 1. Interview Questions draft

Question 1: Experiences with Disruptions - Can you describe one or more specific situations where you experienced a disruption in the supply chain? Probes: - What happened step by step? - How did you initially respond? - What was the most difficult aspect to handle? - How was the situation ultimately resolved?
Question 2: How they work in practice

- How do you typically work when something unexpected occurs in the supply chain?

Probes:

- Are there any formal routines, or do you handle situations case by case?
- Who makes decisions in such situations?
- How are decisions made when something unexpected happens?
- Is the process structured or more spontaneous?
- How quickly do you need to act?

Question 3: Systems and Technology

- Do you use any digital systems or technological tools in your supply chain work (e.g., ERP systems, planning systems, or Excel)?

Probes:

- How are these used in daily operations?
- Do they help you manage disruptions, and if so, how?
- Is there anything that works particularly well or less well with these systems?

Question 4: Internal Capabilities

- Which organisational capabilities do you consider most important for handling disruptions?

Probes:

- How does flexibility influence how you manage disruptions?
- How do experience or "gut feeling" affect decision-making?
- How do you collaborate internally in such situations?
- Have you learned anything from previous disruptions?

Question 5: Resource Constraints

- Do limited resources affect how you handle disruptions?

Probes:

- Is there anything you would like to do but are unable to?
- How do you prioritise when not everything can be resolved?
- How does being a small organisation affect your way of working?

Question 6: Suppliers & Relationship

- What role do your suppliers play when problems arise?

Probes:

- Do you have close relationships with certain suppliers?
- Can you influence them, or are you dependent on them?
- How do you communicate during disruptions?
- Have relationships helped you solve problems?
- Please provide an example of when someone external helped.

Question 7: Proactiveness & future

- Do you take any measures to prevent future disruptions?

Probes:

- Can you give an example of something you changed after a previous disruption?
- What would you like to improve moving forward?
- What do you think would make you more resilient?
- Is this something you actively work on, or only address when needed?

Question 8: Reflection

- If you reflect on how you currently work, what do you think works well and what works less well?

Appendix 2. Interview Findings

Main Theme	Sub-code	Interpretation	Quote / Content
Organisational capabilities	Flexibility in working methods	Work is handled situationally rather than through formal routines	"it is handled case by case" (Respondent 5)
Organisational capabilities	Informal decision-making	Decisions are based on experience and intuition rather than structured processes	"we just keep going and deal with things as they come" (Respondent 4)

Organisational capabilities	Internal communication	Lack of structured communication affects planning and coordination	Need for clearer communication between office and production
Organisational capabilities	Leadership and coordination	Centralised decision-making enables speed but creates dependency	"Top management makes the decisions"
Organisational capabilities	Dependency on individuals	Knowledge and responsibilities are concentrated to few individuals	When employees leave, knowledge is lost, as certain knowledge is held by individuals
Organisational capabilities	Vulnerability to absence	Operations are disrupted when key individuals are absent	"if someone is absent due to illness, the entire chain can come to a halt" (Respondent 5)
Organisational capabilities	Organisational learning	Learning occurs informally through experience rather than structured processes	Backup suppliers established after disruptions
Organisational capabilities	Lack of routines	Work is not standardised	"we have no established routines"
Organisational capabilities	Lack of structure	Missing planning processes and coordination mechanisms	No regular meetings or follow-ups
Organisational capabilities	Unclear responsibilities	Roles and responsibilities are not clearly defined	Unclear ownership in purchasing/planning
Organisational capabilities	Knowledge concentration	Knowledge is not systematically shared	Competence tied to individuals
Organisational capabilities	Operational priority	Daily work limits improvement initiatives	No time for development work
Supplier relationships	Supplier dependency	Reliance on suppliers creates vulnerability	Limited influence over suppliers
Supplier relationships	Close supplier relationships	Strong relationships enable flexibility and problem-solving	"it's important to have a good relationship with critical suppliers" (Respondent 7)
Supplier relationships	Early information sharing	Suppliers provide early warnings about disruptions	Supplier informed about shortages during COVID-19 (Respondent 1)
Supplier relationships	Backup suppliers	Alternative suppliers used to reduce risk	"we have backup suppliers"
Resource constraints	Capital constraints	Financial limitations restrict inventory decisions	"we do not want to tie up capital" (Respondent 6)
Resource constraints	Time constraints	Lack of time limits improvement and development work	"everything is managed on a day-to-day and hour-to-hour basis" (Respondent 5)
Resource constraints	Staffing limitations	Small workforce limits flexibility and redundancy	Difficulty covering absences
Resource constraints	Reactive work patterns	Daily operations prioritised over long-term planning	Focus on short-term problem-solving

Technology as support	ERP system usage	ERP system is central but underutilised	“Monitor is only used to around 30%” (Respondent 7)
Technology as support	Lack of system integration	Incomplete data reduces system effectiveness	Missing articles in BOM
Technology as support	Lack of training	Employees lack formal training in systems	Learning by doing / colleagues
Technology as support	Workarounds	Manual solutions complement system use	Use of Excel for calculations
Technology as support	Resistance to change	Reluctance to adopt new ways of working	Hesitation towards new systems/processes